

Coffee, tea, and the risk of hip fracture: a meta-analysis.

The present meta-analysis shows no clear association between coffee consumption and the risk of hip fractures. There was a nonlinear association between tea consumption and the risk of hip fracture. Compared to no tea consumption, drinking 1-4 cups of tea daily was associated with a lower risk of hip fracture.

INTRODUCTION: Prospective cohort and case-control studies have suggested that coffee and tea consumption may be associated with the risk of hip fracture; the results have, however, been inconsistent. We conducted a meta-analysis to assess the association between coffee and tea consumption and the risk of hip fracture.

METHODS: We performed systematic searches using MEDLINE, EMBASE, and OVID until February 20, 2013, without limits of language or publication year. Relative risks (RRs) with 95% confidence intervals (CI) were derived using random-effects models throughout all analyses. We conducted categorical, dose-response, heterogeneity, publication bias, and subgroup analyses.

RESULTS: Our study was based on 195,992 individuals with 9,958 cases of hip fractures from 14 studies, including six cohort and eight case-control studies. The pooled RRs of hip fractures for the highest vs. the lowest categories of coffee and tea consumption were 0.94 (95% CI 0.71-1.17) and 0.84 (95% CI 0.66-1.02), respectively. For the dose-response analysis, we found evidence of a nonlinear association between tea consumption and the risk of hip fracture ($p(\text{nonlinearity}) < 0.01$). Compared to no tea consumption, 1-4 cups of tea per day may reduce the risk of hip fracture by 28% (0.72; 95% CI 0.56-0.88 for 1-2 cups/day), 37% (0.63; 95% CI 0.32-0.94 for 2-3 cups/day), and 21% (0.79; 95% CI 0.62-0.96 for 3-4 cups/day).

CONCLUSIONS: We found no significant association between coffee consumption and the risk of hip fracture. A nonlinear association emerged between tea consumption and the risk of hip fracture; individuals drinking 1-4 cups of tea per day exhibited a lower risk of hip fractures than those who drank no tea. The association between 5 daily cups of tea, or more, and hip fracture risk should be investigated.